

THERMO GUARD

AI-Driven Biometeorological Early Warning & Heat Disaster Mitigation System

Official Technical Walkthrough, System Architecture & Operational Manual | Version 2.4.0 Production

Specification Parameter	System Value / Operational Status
Target Geographic Jurisdiction	Pan-India (700+ Districts across all 28 States & 8 Union Territories)
Biometeorological Standards	ISO 7243 (WBGT), NOAA Heat Index (Rothfusz), Canadian Humidex, UTCI
Governance & SOP Compliance	National Disaster Management Authority (NDMA) & IMD Heat Action Plan (HAP)
Predictive Artificial Intelligence	Ensemble Machine Learning Regressors with Explainable AI (XAI) Factor Attribution
Automated Civil Reporting	1-Click NDMA Daily Situation Report (SITREP) Cryptographic PDF Generation
Client Architecture & Support	Universal Responsive PWA (React 19, TypeScript, Tailwind) - 320px to 4K Displays
Quality Assurance & Testing	100% Pass (29/29 Automated Security, Health Risk, Alert & SITREP Tests)

1. Executive Overview & Problem Context

Why single-parameter air temperature forecasting fails to protect human lives during extreme heatwaves.

The Critical Flaw of Conventional Weather Alerts

- Humidity Evaporation Failure: At 38°C with 75% relative humidity, human sweat cannot evaporate. The physiological apparent heat exceeds 52°C, inducing rapid cardiovascular collapse and fatal heat stroke.
- Solar Radiative Burden: Direct sunlight and asphalt/concrete thermal radiation add 8°C to 15°C of effective biological thermal burden above standard shaded thermometer readings.
- Urban Heat Island & Nocturnal Trapping: Metropolitan concrete retain thermal energy, preventing nocturnal cooling below 28-30°C. This denies the human body its mandatory cardiovascular recuperation window.
- Vulnerable Labor Force: Over 75% of India's workforce is engaged in exposed informal labor (agriculture, construction, street vendors, delivery) lacking real-time, personalized protective guidance.

The THERMO GUARD Solution

THERMO GUARD transforms passive meteorology into active, life-saving disaster intervention. It unifies biometeorological indices (WBGT, Humidex, Heat Index) with explainable machine learning predictions, population demographic weighting, and an automated NDMA situation room to empower both grassroots citizens and disaster commanders.

2. System Architecture & Mathematical Engine

End-to-end telemetry ingestion, thermodynamic modeling, and explainable AI risk scoring.

Multi-Tier Telemetry Ingestion Pipeline

- Primary Weather Engine: Real-time high-resolution meteorological telemetry ingested via Open-Meteo Ensemble and IMD gridded observation pipelines.
- Core Tracked Variables: 2-meter ambient dry-bulb temperature, relative humidity, dewpoint temperature, surface wind speed, surface solar irradiance, UV index, and elevation metrics.
- Geospatial Harmonization: Pan-India dataset covering all 700+ districts with verified geographical centroids, historical climate baselines, and Census population density distributions.

Index Name	Mathematical Formulation	Clinical / Occupational Application
Wet Bulb Globe Temperature (WBGT) [ISO 7243]	$WBGT_{out} = 0.7 \cdot T_w + 0.2 \cdot T_g + 0.1 \cdot T_a$ (T_w derived via Stull iterative psychrometry; T_g models direct solar & radiative balance)	Gold standard for industrial, agricultural, and construction work-rest cycles and labor safety.
Canadian Humidex Scale	$Humidex = T_a + (5/9) \cdot (e - 10)$ where e = actual vapor pressure in hPa $e = 6.112 \cdot 10^{[(7.5 \cdot T_a)/(237.7 + T_a)]}$ $\cdot (RH/100)$	Quantifies atmospheric stuffiness and physiological discomfort due to high ambient moisture.
NOAA Heat Index (Rothfus Regression)	$HI = c_1 + c_2 \cdot T + c_3 \cdot RH + c_4 \cdot T \cdot RH + \dots$ (9-term polynomial with dry & humid boundary condition adjustments)	Apparent temperature perception scale utilized for general civil and residential heat alerts.
Heat Thermal Stress Score (HTSS) [0-100]	$HTSS = w_1 \cdot WBGT_{norm} + w_2 \cdot Humidex_{norm} + w_3 \cdot UV_{norm} + w_4 \cdot NightTrap_{norm} + w_5 \cdot HVI$	THERMO GUARD proprietary multi-factor composite risk index powering color-coded alert matrix.

Explainable Artificial Intelligence (XAI) & Predictive Risk Engine

- Supervised Ensemble Predictive Regressor: Evaluates 7-day future heat trajectories by assessing diurnal temperature delta, rolling 3-day thermal accumulation, and surface wind stagnation.
- Transparent Factor Attribution: Replaces black-box uncertainty with verifiable factor contributions. Administrators and citizens see exact percentage drivers:
 - Ambient Heat Load: ~35-45% | Relative Humidity Moisture: ~25-35%
 - Nocturnal Trapping Effect: ~15-20% | UV Solar Radiation: ~8-12%
- Public Model Transparency: Embedded audit panels expose live training confidence, error margins (RMSE), and validation benchmarks directly to decision-makers.

3. Citizen Early Warning & Health Portal Walkthrough

Empowering vulnerable individuals and outdoor workers with actionable, hyper-localized safety intelligence.

A. Live Biometeorological Thermal Stress Gauge

- Dynamic 5-Stage Color Spectrum: Instant visual representation of biological danger:
[SAFE - Green (<30)] | [CAUTION - Yellow (30-34)] | [EXTREME CAUTION - Amber (35-39)]
[DANGER - Orange (40-44)] | [EXTREME DANGER - Red (45+)]
- Dual-Metric Transparency: Displays both ambient thermometer reading and true felt apparent temperature with real-time humidity, wind speed, and UV index status.
- Micro-Animations & Hardware Acceleration: Smooth pointer movements and high-contrast numerical readouts ensure rapid comprehension under direct outdoor sunlight.

B. Dynamic Work-Rest & Hydration Scheduler

- NIOSH & OSHA Compliant Work-Rest Planning: Calculates precise work-rest ratios tailored to environmental heat stress:
 - Low Stress: Continuous work with 250ml water/hr.
 - Moderate Stress: 45 min work / 15 min shaded rest with 500ml water/hr.
 - High Stress: 30 min work / 30 min shaded rest with 750ml water/hr.
 - Extreme Stress: Immediate suspension of non-essential outdoor physical labor.
- Activity Intensity Selector: Customizes recommendations across Sedentary, Moderate, Heavy, and Extreme physical workloads.

C. Personalized Vulnerability Profiling

- Demographic Risk Tuning: Allows users to configure profiles for Children, Adults, Elderly individuals, or Outdoor Workers.
- Medical Comorbidity Awareness: Provides tailored cautionary advisories for users with hypertension, cardiovascular conditions, diabetes, respiratory ailments, or pregnancy.
- Local Storage Privacy: All personal health configurations are stored strictly on-device via localStorage—zero health tracking or personal data harvesting.

D. Emergency Heat Alert & SOS Modal

- Automated Crisis Overlay: Triggers immediately when local HTSS crosses critical threshold boundaries.
- Clinical Symptom Differential: Clear side-by-side guide distinguishing Heat Exhaustion (profuse sweating, pale skin, nausea) from Heat Stroke (dry hot skin, confusion, loss of consciousness).
- Emergency Medical Guidance: One-touch emergency dialers, nearest hospital cooling centers, and critical life-saving first-aid protocols (rapid ice cooling, hydration instructions).

4. Government & Disaster Authority Command Matrix

Operational command portal for NDMA, SDMA, and District Collectors (DDMA) to enforce Heat Action Plans.

A. 700+ District Real-Time Heat Action Matrix

- Pan-India Administrative Oversight: Comprehensive real-time monitoring of all districts with live state filtering, search-as-you-type, and severity classification tabs.
- High-Contrast Sticky Headers: Table headers remain permanently anchored during vertical scrolling for zero disorientation during rapid operational reviews.
- Mobile Horizontal Swipe Indicators: Native touch gesture indicators guide operators on compact screens with zero truncated metrics or split numerical units.
- Direct Actionable Columns: State, District, Composite HTSS, Ambient Temperature, Apparent Heat, Humidity, Wind Speed, Population at Risk, and Alert Status.

B. 1-Click Automated NDMA Daily Situation Report (SITREP) Engine

- Elimination of Reporting Latency: Replaces manual, multi-hour administrative report compilation with instantaneous, standardized vector PDF generation.
- Official Cryptographic Layout: Standardized NDMA Heat Action Plan formatting complete with national command headers, officer credential blocks, and cryptographic verification stamps.
- Executive Threat Summary Cards: Highlights total high-risk districts, affected population tallies, and state-wide severity distributions.
- Statutory Civil Directives: Generates legally binding, sector-specific directives covering cooling shelters, municipal water supply, hospital bed reserves, and labor schedule modifications.

NDMA Alert Stage	Thermal Threshold	Mandatory District Civil Action Checklist
White (Normal)	HTSS < 30	Routine public weather monitoring; seasonal preparedness review.
Yellow (Watch)	HTSS 30 - 34	Issue public health advisories; inspect public drinking water points.
Orange (Alert)	HTSS 35 - 39	Open public cooling centers; adjust outdoor work hours; stock hospital ORS supplies.
Red (Warning)	HTSS 40+	Activate emergency ops; deploy mobile water tankers; suspend outdoor labor from 11AM-4PM; reserve emergency heat-stroke wards.

C. Resource Allocation & Shelter Logistics Matrix

- Proactive Infrastructure Dispatch: Maps district heat vulnerability against municipal infrastructure, identifying critical zones for emergency water kiosk deployment.
- Hospital Bed Availability Tracking: Provides health departments with predictive caseload estimates to reserve air-conditioned cooling wards and IV rehydration fluids.

5. Technical Rigor, Quality Assurance & Future Roadmap

Software architecture standards, automated test verification, security posture, and roadmap.

Public Education & Simulation Lab (Ilearn)

- Interactive Humidex & Heat Index Calculators: Allows citizens, students, and health workers to dynamically manipulate temperature and humidity sliders to visualize physiological effects.
- Comprehensive Indian District Vulnerability Atlas: Educates users on why coastal vs. arid vs. river-basin districts exhibit radically distinct heat vulnerability profiles.
- Community First-Aid Masterclass: Step-by-step illustrated emergency procedures for managing heat cramps, heat exhaustion, and preventing fatal heat stroke.

Engineering Rigor & Quality Verification

- 100% Automated Test Pass Rate (29/29 Tests Verified):
 - Security & Sanitization (9 Tests): Guaranteed zero XSS vulnerabilities, strict DOM sanitization, and safe API parameter encoding.
 - Health Risk Regressors (8 Tests): Mathematical boundary validation for all biometeorological algorithms across extreme climate extremes.
 - Alert Engine (8 Tests): Validates deterministic threshold triggering for Yellow, Orange, and Red NDMA stages.
 - SITREP Exporter (4 Tests): Cryptographic PDF rendering, binary integrity, and zero text truncation verification.
- Zero TypeScript Compilation Errors: Fully compliant across Vite module compiler (`tsc -b`) and root serverless configuration (`tsc --noEmit`).

Device Accessibility & Universal Responsive Design

- Ultra-Compact Mobile (320px–390px): Optimized margins, non-clipping location titles, wrapped badge clusters, and horizontal swipe banners.
- Tablets (768px): Flexible split-column overview cards, responsive touch gauges, and sticky table layouts.
- Command Center Monitors (1080p–4K): Multi-column incident matrices, expanded GIS maps, and live real-time telemetry streaming.

Future Scalability Roadmap

- Phase 2 - IoT Microclimate Network: Ingestion of real-time telemetry from municipal LoRaWAN smart weather nodes and agricultural field sensors.
- Phase 3 - Voice & SMS Fallback Dissemination: Automated regional language voice calls (IVR) and SMS broadcasts for non-smartphone users in rural agricultural belts.
- Phase 4 - Satellite Remote Sensing: Integration of ISRO INSAT-3D/3DR land surface temperature (LST) imagery for automated high-resolution Urban Heat Island (UHI) mapping.